```
import numpy as np
```

```
arr = np.array([10,20,30,40,50])
print(arr)
```

```
[10 20 30 40 50]
```

Slicing Syntax

array[start:end]

```
arr = np.array([10,20,30,40,50])
print(arr[1:4])
print(arr[:3])
print(arr[2:])
print(arr[-3:])
```

```
[20 30 40]
[10 20 30]
[30 40 50]
[30 40 50]
```

Numpy Functions

```
# arrange()
arr = np.arange(1,6)
print(arr)
```

```
[1 2 3 4 5]
```

```
#Zeros()
print(np.zeros(5))
```

```
[0. 0. 0. 0. 0.]
```

```
#ones()
print(np.ones(5))
```

```
[1. 1. 1. 1. 1.]
```

```
# sum(), mean(),max(),min()
marks = np.array([70,80,90,60])

print("sum:",np.sum(marks))
print("Average:",np.mean(marks))
print("Max:",np.max(marks))
print("min:",np.min(marks))
```

```
sum: 300
Average: 75.0
Max: 90
min: 60
```

2D Numpy Array

```
arr = np.array([
    [10,20,30],
    [40,50,60],
    [70,80,90]
])
print(arr)
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]]
```

```
print(arr.shape)
```

```
(3, 3)
```

2d Array accessing Elements

array[row_index,column_index]

```
print(arr[0,1])
```

```
20
```

```
print(arr[2,1])
```

```
80
```

Double-click (or enter) to edit

```
arr = np.array([
    [10,20,30,11],
    [40,50,60,22],
    [70,80,90,33]
])
print(arr)
```

```
[[10 20 30 11]
 [40 50 60 22]
 [70 80 90 33]]
```

```
print(arr+10)
```

```
import numpy as np
arr = np.array([
    [10,20,30,11],
    [40,50,60,22],
    [70,80,90,33]
])
print(arr[1,2:], arr[2])
```

```
[60 22] [70 80 90 33]
```

```
import numpy as np
arr = np.array([
    [10,20,30,11],
    [40,50,60,22],
    [70,80,90,33]
])
print(arr+100)
```

```
[[110 120 130 111]
 [140 150 160 122]
 [170 180 190 133]]
```

```
import numpy as np
arr = np.array([
    [10,20,30,11],
    [40,50,60,22],
    [70,80,90,33]
])
print(arr*2)
```

```
[[ 20  40  60  22]
 [ 80 100 120  44]
 [140 160 180  66]]
```

```
import numpy as np
arr = np.array([1,4,9,16,25])
print(np.sqrt(arr))
print(np.power(arr,2))
```

```
[1. 2. 3. 4. 5.]
[  1  16  81 256 625]
```

```
import numpy as np
arr = np.array([1.4,2.4,3.4,4.4])
print(np.round(arr))
```

```
[1. 2. 3. 4.]
```

```
import numpy as np
arr = np.array([-1,-2,-4])
print(np.absolute(arr))
```

```
[1 2 4]
```

Operators

```
#comparison oprators
import numpy as np
marks = np.array([10,77,99,567,24])
print(marks>80)
```

```
[False False  True  True False]
```

```
# logical operators
import numpy as np
marks = np.array([70,80,90,60])
print((marks>40) & (marks<70))
print((marks>40) & (marks<70))
```

```
[False False False  True]
[False False False  True]
```

```
#boolean indexing
import numpy as np
marks = np.array([70,80,90,60])
passed = marks[marks>70]
print(passed)
```

```
[80 90]
```

```
# Q. analyze marks and find average,
import numpy as np
marks = np.array([70,80,90,60,25,10])
passed = marks[marks>40]
failed = marks[marks<40]
print(passed)
print(f"average of all students : {np.mean(marks)}")
print(f"average of passed students : {np.mean(passed)}")
print(f"Highest scorer from passed students : {np.max(passed)}")
print(f"Lowest scorer from passed students : {np.min(passed)}")
print(f"average of failed students : {np.mean(failed)}")
```

```
[70 80 90 60]
average of all students : 55.833333333333336
average of passed students : 75.0
Highest scorer from passed students : 90
Lowest scorer from passed students : 60
average of failed students : 17.5
```